Boshi Wang

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Education

ShanghaiTech University Shanghai, China

B.E. in Computer Science and Technology Sep. 2016 - Present

- **GPA**: 3.76/4.0 (Rank 5/116)
- **Graduate-level Coursework**: Convex Optimization, Matrix Analysis, Introduction to Algorithmic Game Theory, Subspace Learning (Fall 2019)
- **Programming Skills**: C/C++, Python, Matlab, Rust, MIPS assembly
- **Tools**: CVX, Robot Operating System (ROS), Spark, Tensorflow, Pytorch

Publications/Papers

* denotes equal contribution

- Y. Guo*, **B. Wang***, D. Hughes, M. Lewis, K. Sycara. Designing Context-Sensitive Norm Inverse Reinforcement Learning Framework for Norm-Compliant Autonomous Agents. *Submitted to ICRA'20.* **(Under Review, *Link to Paper*)**
- **B. Wang**. Homomorphic Sensing with Union of Linear Subspaces as Input Space. https://boshi-wang.github.io/documents/HS_with_Union_of_Subspaces.pdf**(Working Paper, *Link to Paper*)**
- Z. Tu, **B. Wang**, T. Duan, and F. Gao. Pattern-learning-based Noise Elimination Algorithm in Photoacoustic Sensing and Imaging. In *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'18)*, Honolulu, HI, USA, on July 17-21, 2018. **(Published, *Link to Paper*)**

Research Experience

Advanced Agent-Robotics Technology Lab, Carnegie Mellon University Pittsburgh, USA

Undergraduate Research Intern, supervised by Prof. *Katia Sycara* Jun 2019 - Sep 2019

- Developed Context-Sensitive Norm Inverse Reinforcement Learning (CNIRL), a computational scalable Inverse-RL framework for developing norm-compliant autonomous agents through learning from demonstrations under large number of environmental contexts. Link to the *Poster* and *Paper*

Manolis Research Group, ShanghaiTech University Shanghai, China

Undergraduate Research Assistant, supervised by Prof. *Manolis C. Tsakiris* Feb 2019 - Present

- Extended Homomorphic Sensing (a recent ML research topic proposed by Prof. Tsakiris in an *ICML'19* paper) into the case where the input space is a union of finitely many linear subspaces instead of a single linear subspace. Link to *Working Paper*
- Currently conducting theoretical analysis on subspace recovery under partially permuted data with Dual Principal Component Pursuit (DPCP), and also on probabilistic analysis of noisy Homomorphic Sensing.

Hybrid Imaging System Laboratory, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Fei Gao* Jan 2018 - Nov 2018

- Developed an algorithm that can learn from historical data about the pattern of a periodic noise in Photoacoustic signals and efficiently perform noise reduction. (Paper accepted by *EMBC'18*, *Link*)
- Conducted research on using denoising autoencoders (DAE) for photoacoustic signal noise reduction. The denoising behavior largely outperforms traditional methods under low signal-to-noise ratio.

Mobile Autonomous Robotic Systems Lab, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Sören Schwertfeger* Jun 2017 - Dec 2017

- Implemented Visual Odometry using Aruco markers in the surrounding environment, and combined it with motion-based odometry (developed by collaborators) via G2O to achieve high localization accuracy. The work contributes to a bigger project in the lab about Simultaneous Localization and Mapping (SLAM).

Teaching Experience

Teaching Assistant of *SI140: Probability and Mathematical Statistics*, ShanghaiTech University.
Course Instructor: Prof. *Ziyu Shao* Shanghai, China

Industry Experience

Chinese Software Developer Network Corporation (CSDN) Beijing, China
Intern of System Developer Jun 2018 - Sep 2018

- Optimized the keyword matching and filtering algorithm of the blog recommendation system.
- Developed an efficient method for clustering search queries of blogs and implemented the method with parallel computing using Spark.

Other Selected Projects

Deep Q-learning based AI agent for Beating the Landlord
Course project of *CS181: Artificial Intelligence* in ShanghaiTech University, Fall 2018

- Developed a Deep Q-learning framework which learns to play a popular 3-player card game in China called "Beating the Landlord" through three agents' self-play. Although the trained agent still can't beat human players, it largely outperforms greedy agents which are used in most online game platforms.

Sparse Representation Based Face Recognition
Course project of *SI131: Linear algebra* in ShanghaiTech University, Fall 2017

- Implemented a face recognition algorithm based on Principle Component Analysis and Sparse Representation based classification. The algorithm achieves 91.4% accuracy on Extended Yale Face Database B.